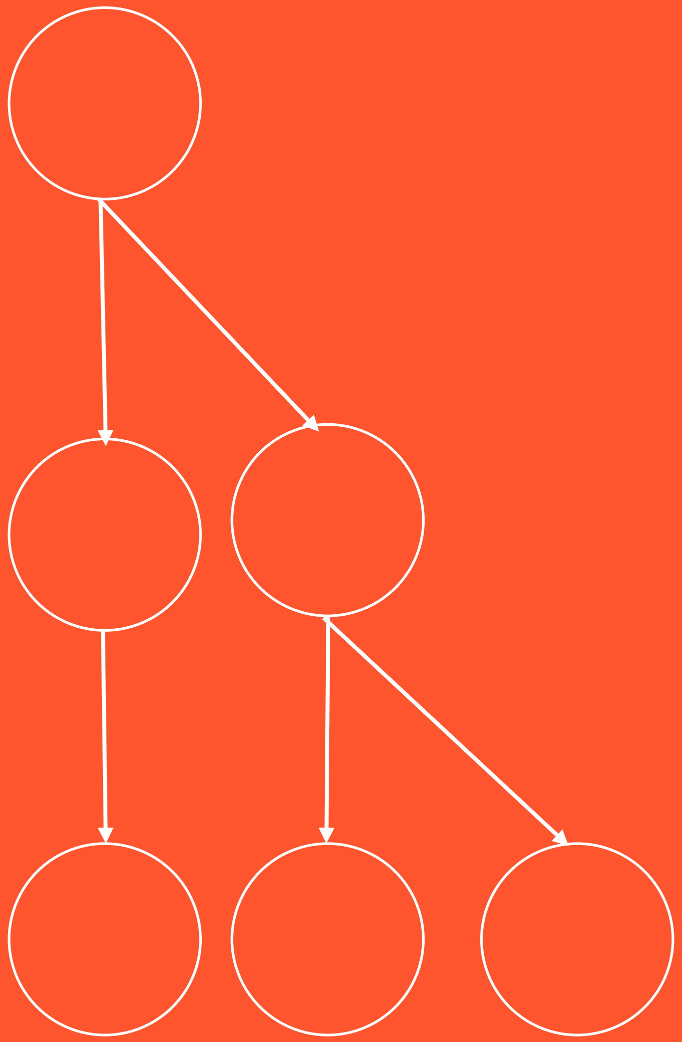


BCOG 100

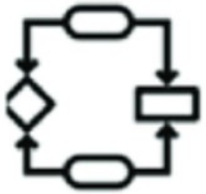
Spatial Cognition

Lecture 2: Spatial Cognition Algorithms and Representations

UNIVERSITY OF ILLINOIS URBANA-CHAMPAIGN

Spatial Cognition Algorithms & Representations

Computation
What does the system do?



Algorithm
How does the system do it?

Implementation
How is the system realized?

Recap & Overview

Why spatial cognition evolved, and what it allowed early vertebrates to do

Today:

- How minds and machines *represent* and *compute* space
- We'll draw on both cognitive psychology and machine learning

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The Computational Challenge

- Given partial, noisy input:
- How can an agent maintain an accurate sense of space?
- Representation + Update = Navigation

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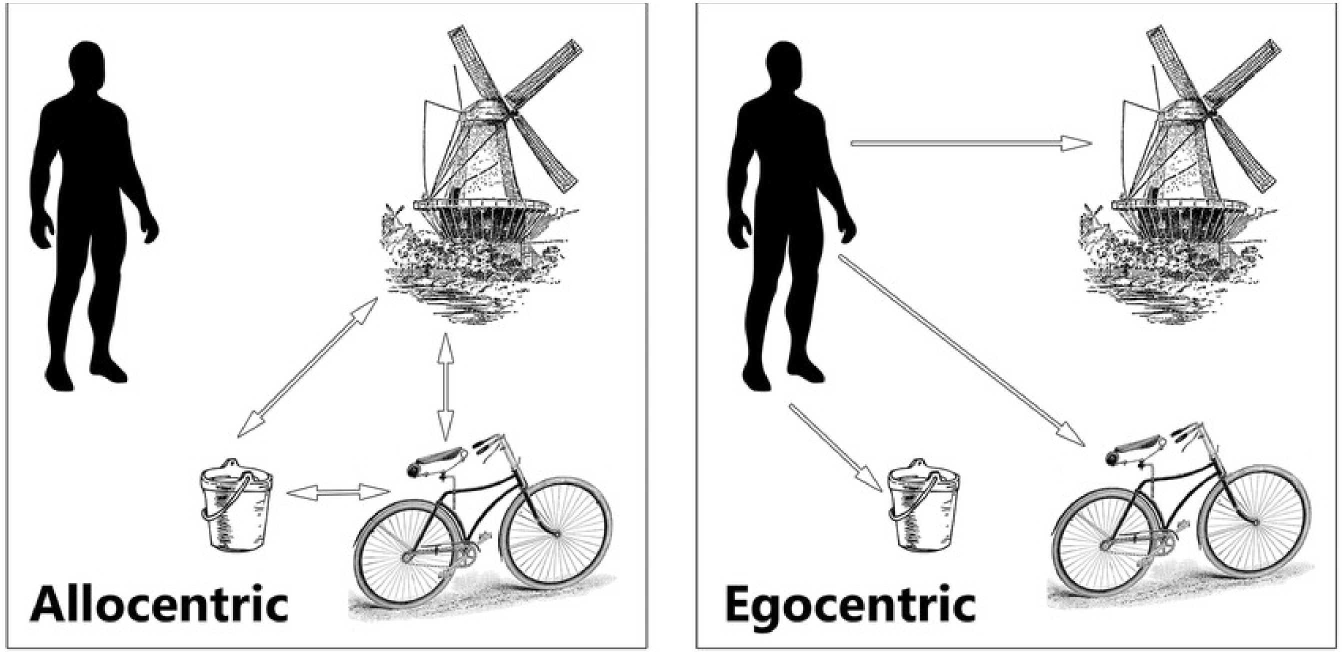

Egocentric and Allocentric Reference Frames

- **Egocentric**: space coded relative to self
- **Allocentric**: space coded relative to environment
- Spatial cognition needs both—and transforms between them

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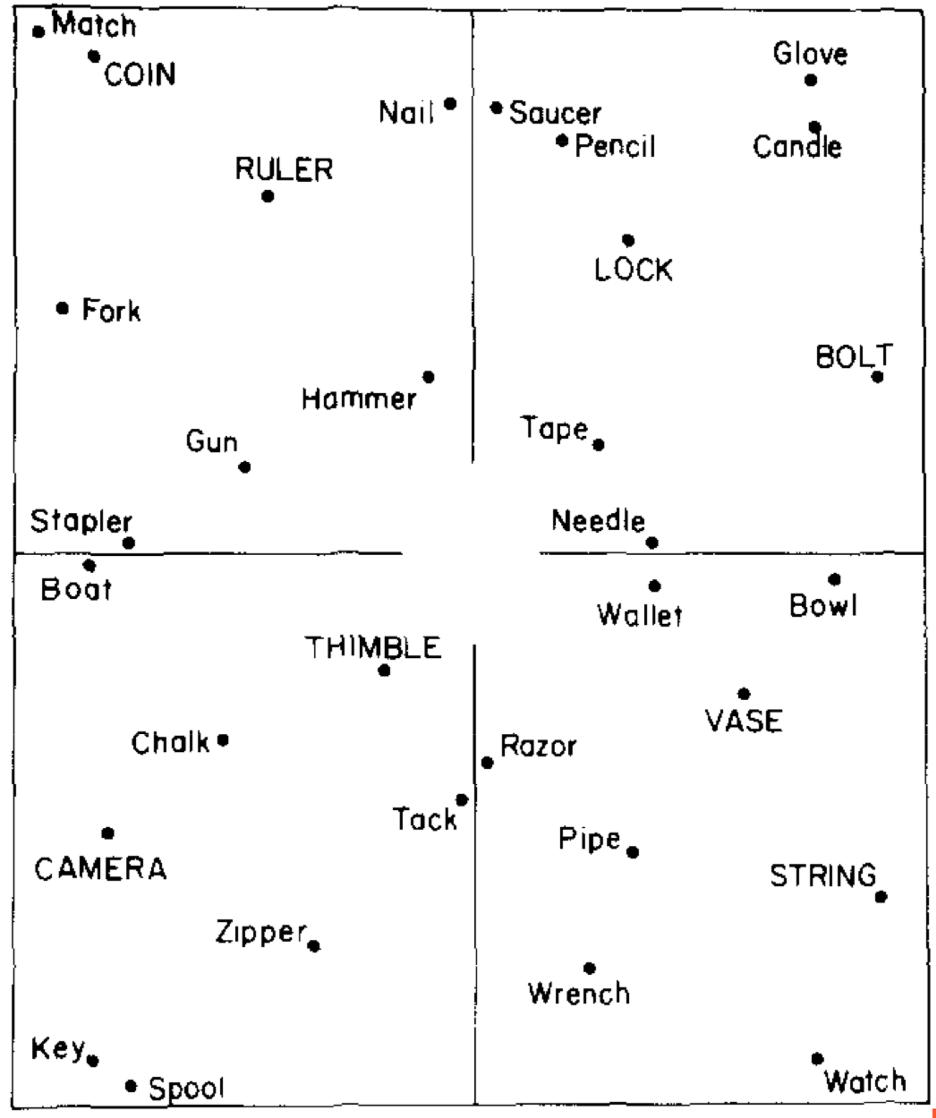

Transforming Between Frames

- Mental maps often align with environmental axes
- Changing perspective slows recall and increases errors
- Reveals the cost of transforming between frames

Reading Check 8-2

Password: map

Question: how did you go about recreating the map?

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What Form Do Mental Representations Take?

- Are mental maps **pictorial** or **propositional**?
- Kosslyn: Mental images are depictive, like inner pictures
- Pylyshyn: Mental images are propositional, abstract descriptions
- How we *represent* space changes what we can *compute*

Pictorial Representation

Propositional Representations

- tree is next to pond
- well and hut are in the SW
- island is shaped like a fat-bottomed ”S”
- beach is in the SE
- E is pretty empty

Spatial Cognition Algorithms & Representations

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Hybrid Representations and Modern Parallels

- Modern evidence: spatial thinking uses both depictive and propositional codes
- Visual and symbolic brain systems cooperate during imagery
- Echoes today: the same questions resurface in debates about LLM “world models”

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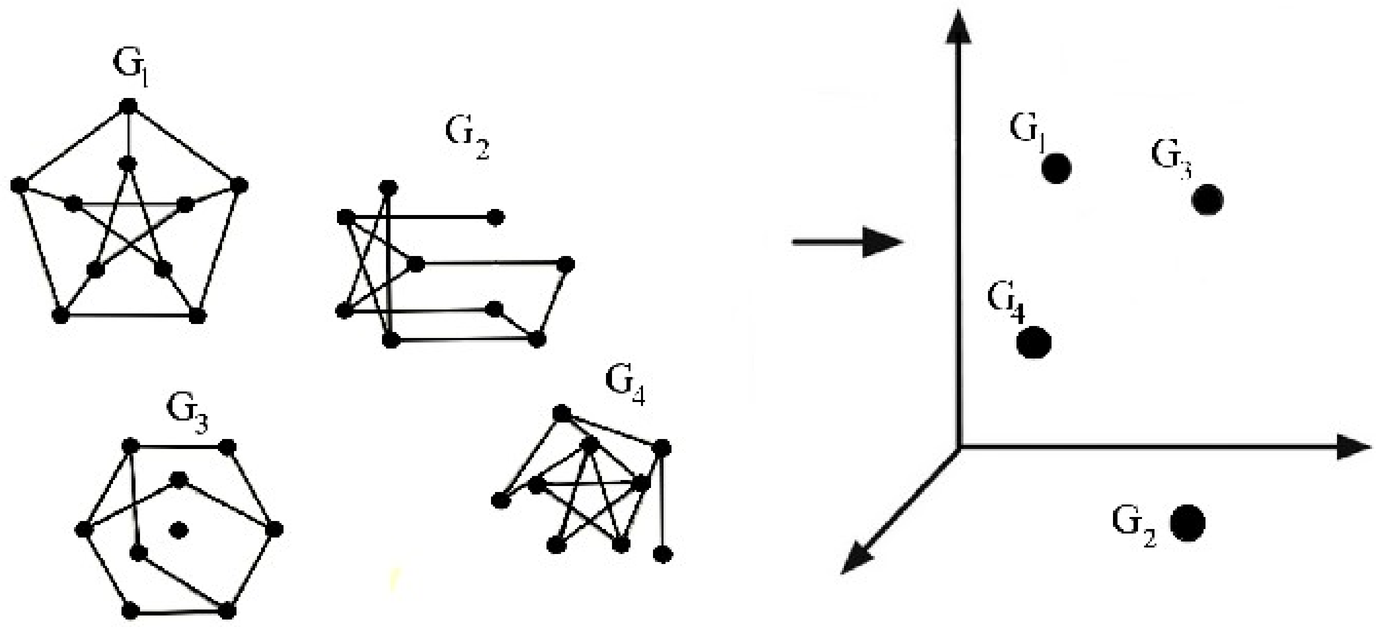
Topological

Metric



Other Candidate Structures

- Metric: precise distances and angles
- Topological: connectivity without scale
- Hierarchical: nested regions and routes

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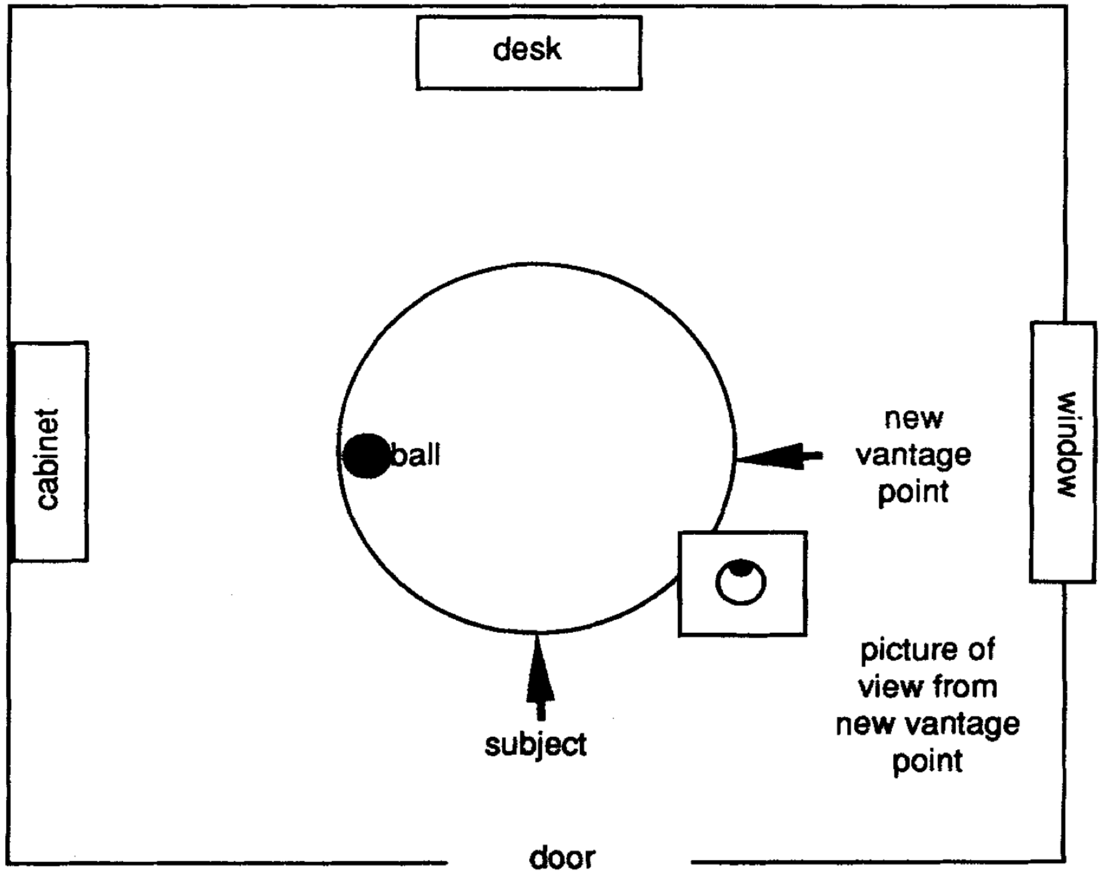

Figure 1. Example of room, array, and picture in a picture-selection task.

Landmarks and Spatial Hierarchies

- Landmarks anchor spatial memory and reduce drift
- Children and adults organize space into nested regions
- Spatial hierarchies explain many everyday navigation errors

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Path Integration and Updating Mechanisms

- Integrating self-motion to estimate position
- Errors accumulate without external cues
- Brains and machines use feedback to recalibrate

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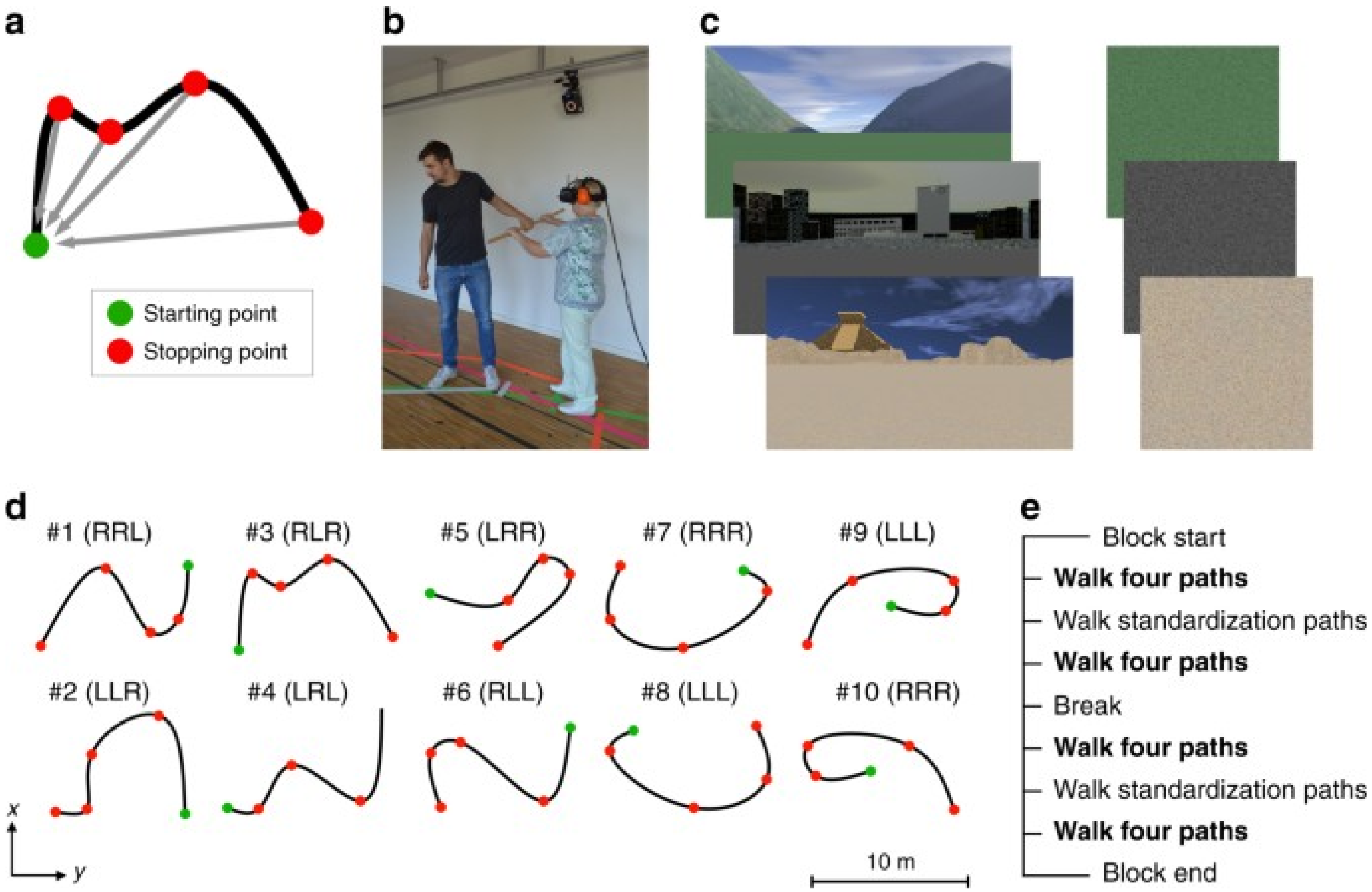

Human and Animal Evidence for Path Integration

- Many species track position by integrating motion
- Errors grow with distance and rotation
- External cues reset and stabilize internal estimates

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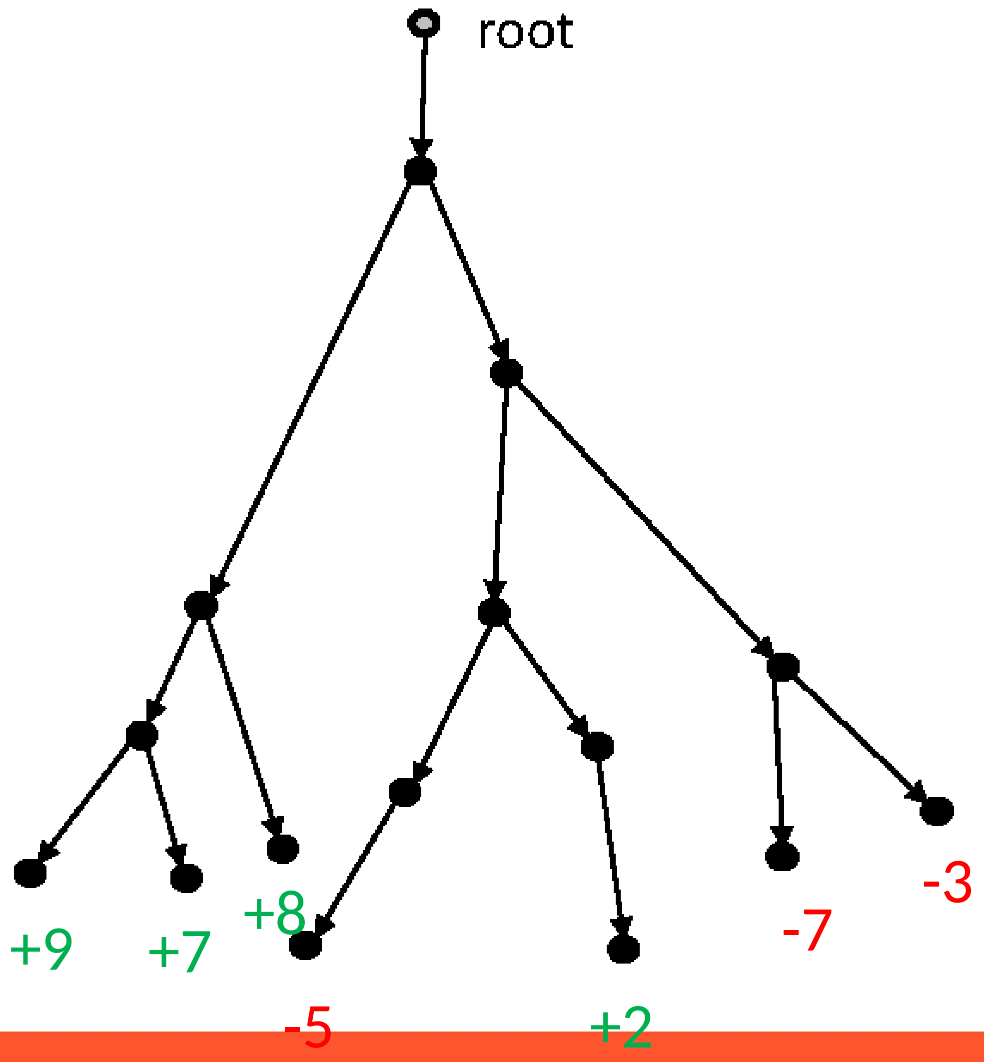

Exploration and the Value and Cost of Novelty

- Once you have a map, the question changes: where next?
- Exploration adds information and reduces uncertainty
- Novelty itself can be intrinsically rewarding

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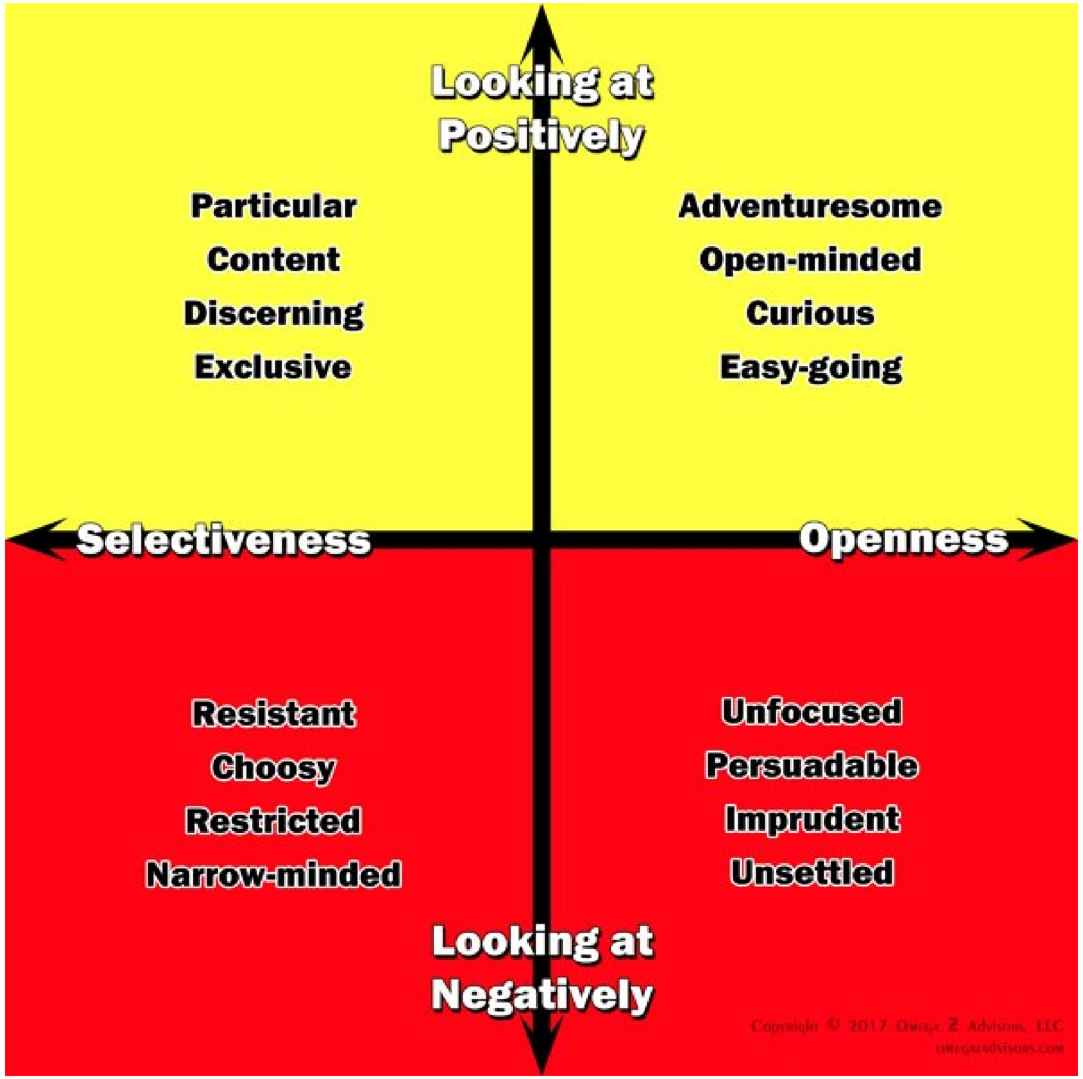

The Explore–Exploit Tradeoff

- Explore: gain new information
- Exploit: use what you already know
- Every organism—and every AI—must balance the two

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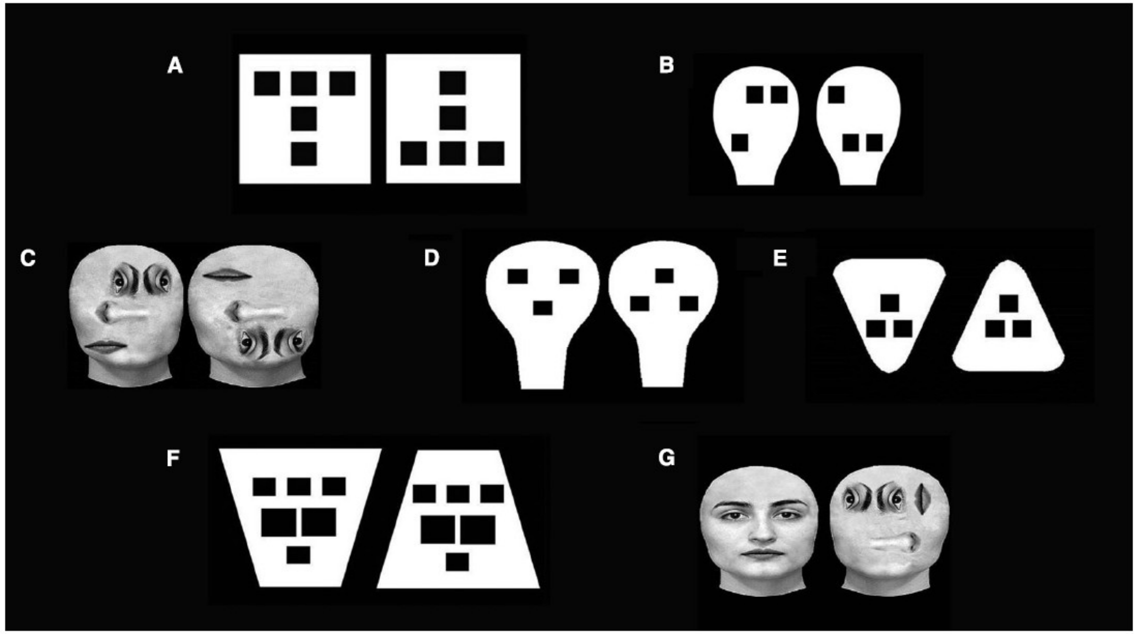

Curiosity and Intrinsic Motivation

Exploration can be rewarding on its own
Curiosity reduces uncertainty and builds knowledge
Play is nature’s way of learning without stakes

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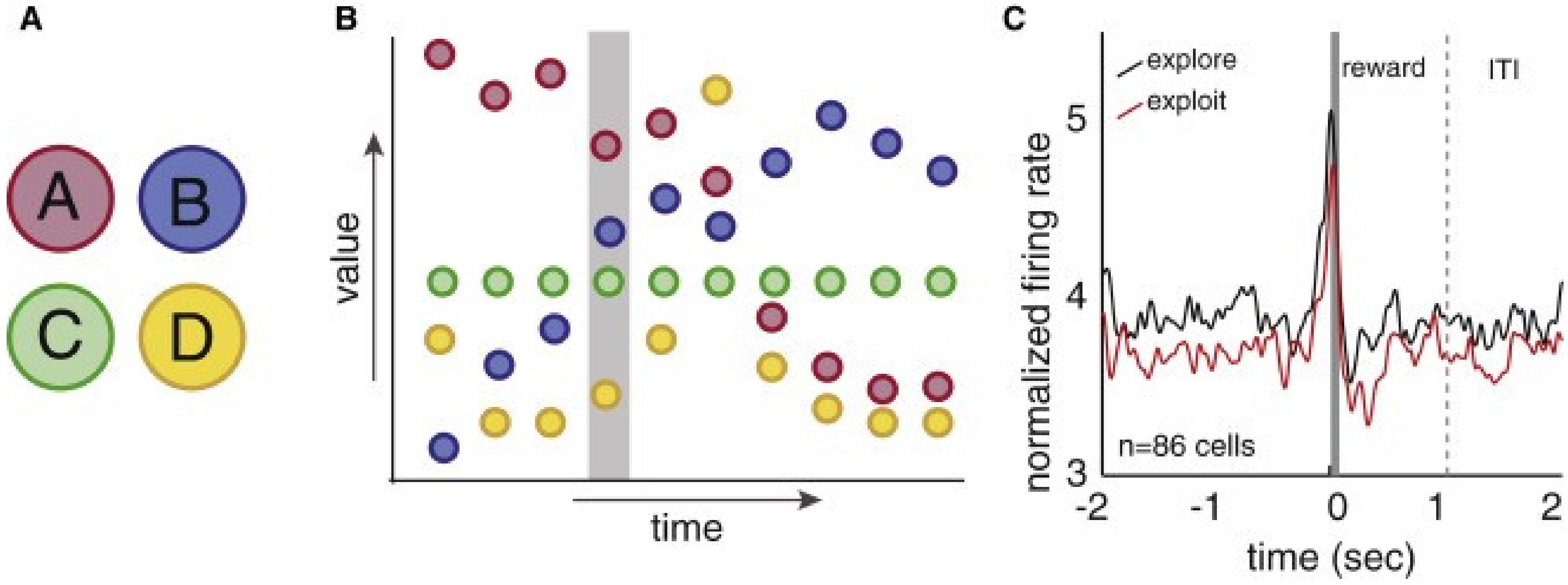

Explore–Exploit in Human and Animal Studies

- Exploration peaks in youth, but persists across life
- Animals and humans explore strategically, not randomly
- Curiosity adapts to uncertainty and potential reward

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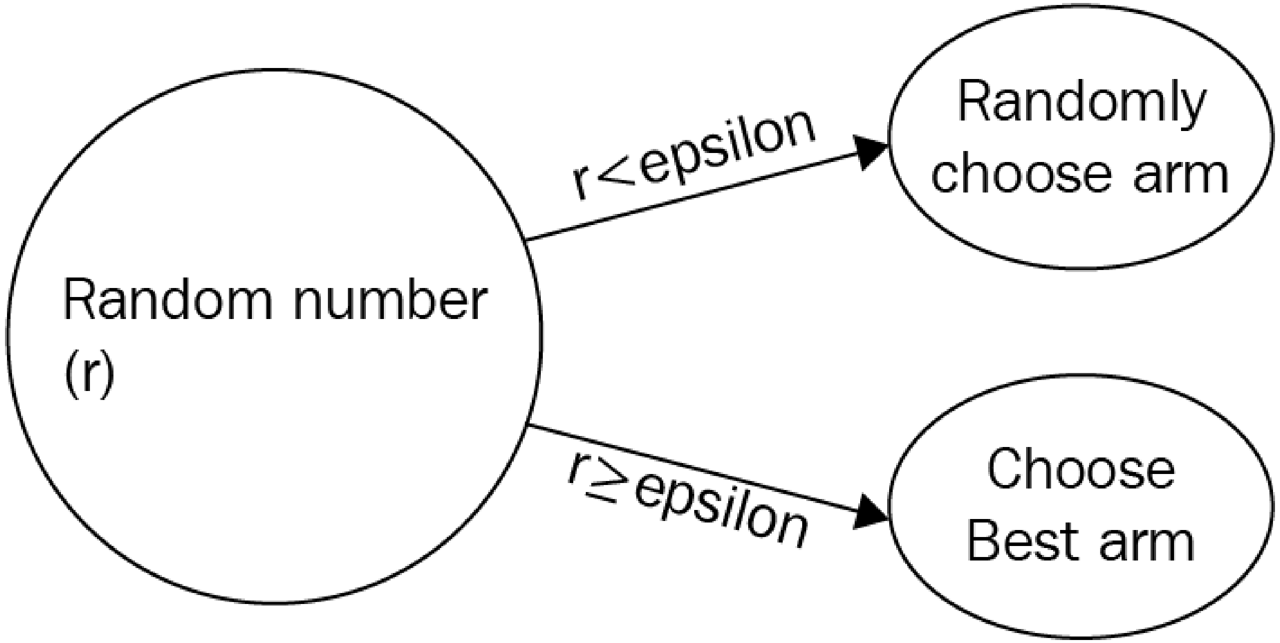

Explore–Exploit in Machine Learning

- AI agents face the same problem
- Too little exploration: they get stuck
- Too much: they never learn what works

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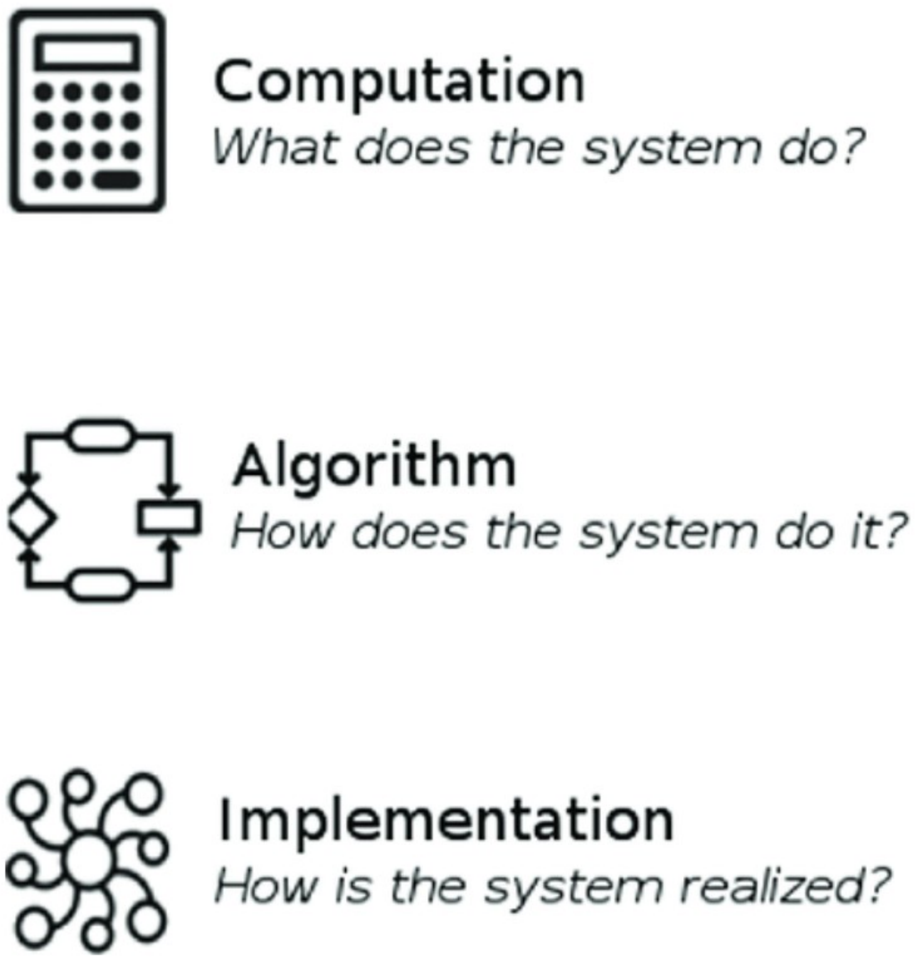

Recap

- There are many kinds of maps we can have
- There are many algorithms for integrating them with input
- Curiosity as a mechanism for building maps

Up Next

How brains implement those computations—From spatial maps to memory systems